

CROWN & BRIDGE

Introduction

Lecture one

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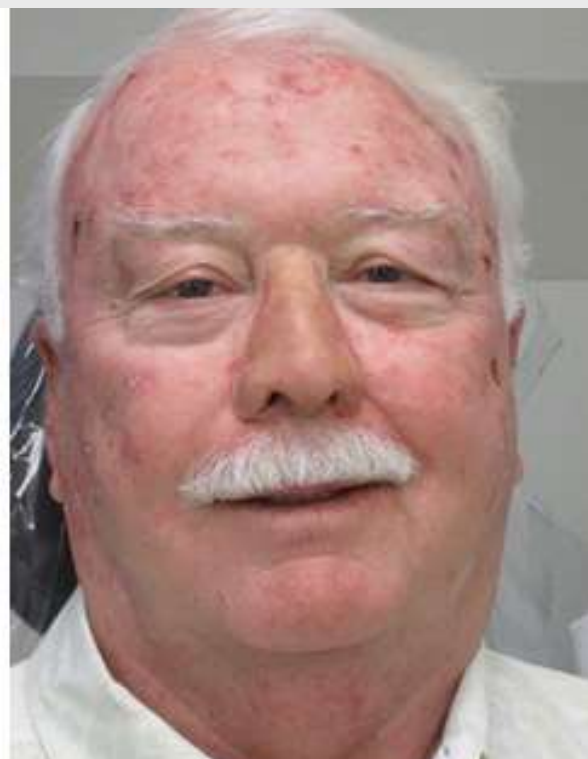
Prosthodontics

It is the dental specialty concerned with the making of artificial replacements for missing parts of the mouth and jaw. It is also named **Prosthetic Dentistry or Prosthodontia.**

It is concerned with manufacturing of the followings

- 1- Complete denture.**
- 2- Removable partial denture.**
- 3- Fixed prosthesis.**
- 4- Artificial oral and facial organs and obturators.**
- 5- Implant related restorations.**





Fixed prosthodontics includes :

- 1- inlays.
- 2- onlays.
- 3- Veneers.
- 4- Crowns.
- 5- Fixed partial dentures (dental bridge).



Definitions :

Fixed prosthesis (crown or Bridge):

It is a branch of dental science (prosthetic) that deals with restoration of damaged teeth with artificial crown and replacing the missing natural teeth by dental prosthesis permanently cemented in place called bridge [Fixed partial denture]



Crown :

It is a fixed extra-coronal artificial restoration of the coronal portion of a natural tooth .It must restore the morphology, contour and the function of the tooth and should protect the remaining tooth structures from farther damage .



Porcelain-fused-to-metal Crown

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The metal portion of a PFM may have value.

Animated-Teeth.com



BEFORE



AFTER





Prepared tooth



Crown placed over prepared tooth

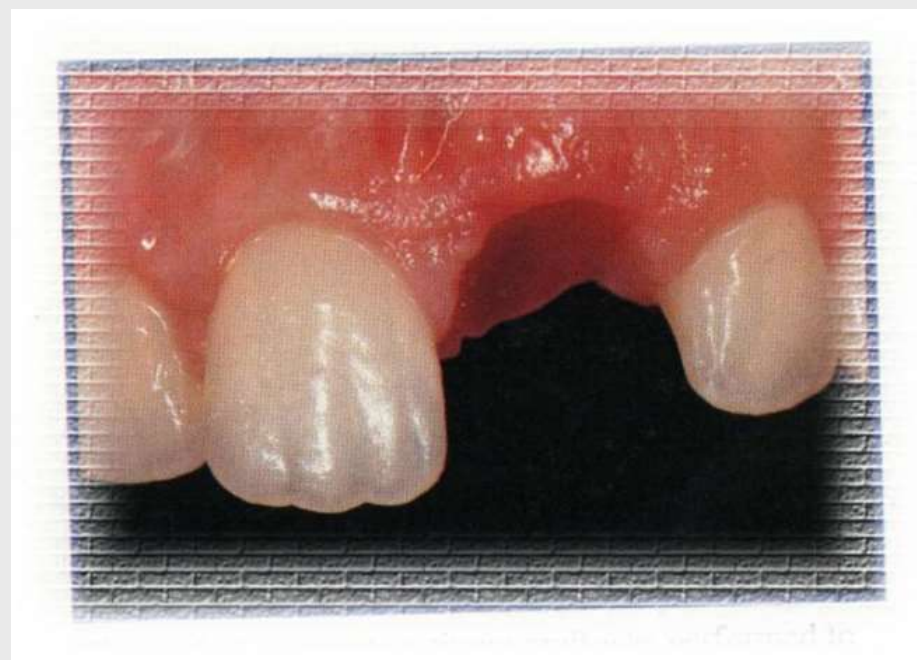


Crown in place



Bridge :

It's a fixed prosthesis constructed to **replace and restore the function and esthetic** of one or more missing natural teeth, and it cannot be removed from the mouth by the patient. It is permanently supported by natural teeth, roots or implants. The tooth that gives support to the bridge is called **abutment tooth**.

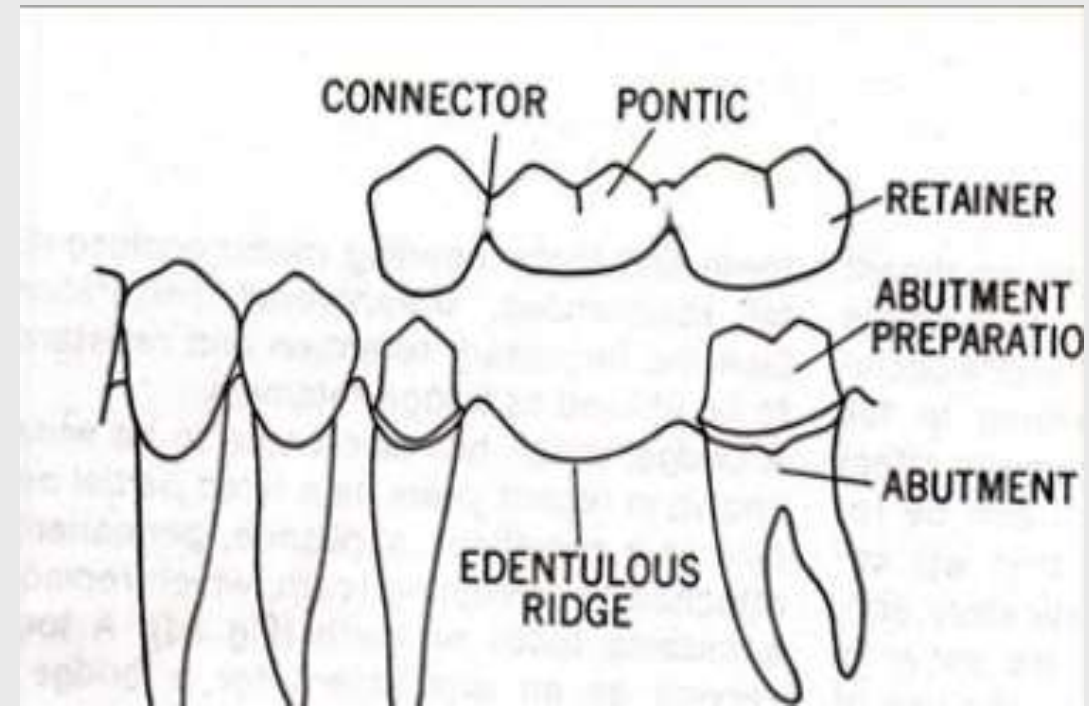




Components of the bridge:

1- Retainer: It is the part that seats over or in the abutment tooth.

2- Pontic : It is that part of the bridge that replaces (represent) the missing tooth or teeth ,Which is the suspended part of the bridge .It occupies the position of the missing natural tooth(ridge).



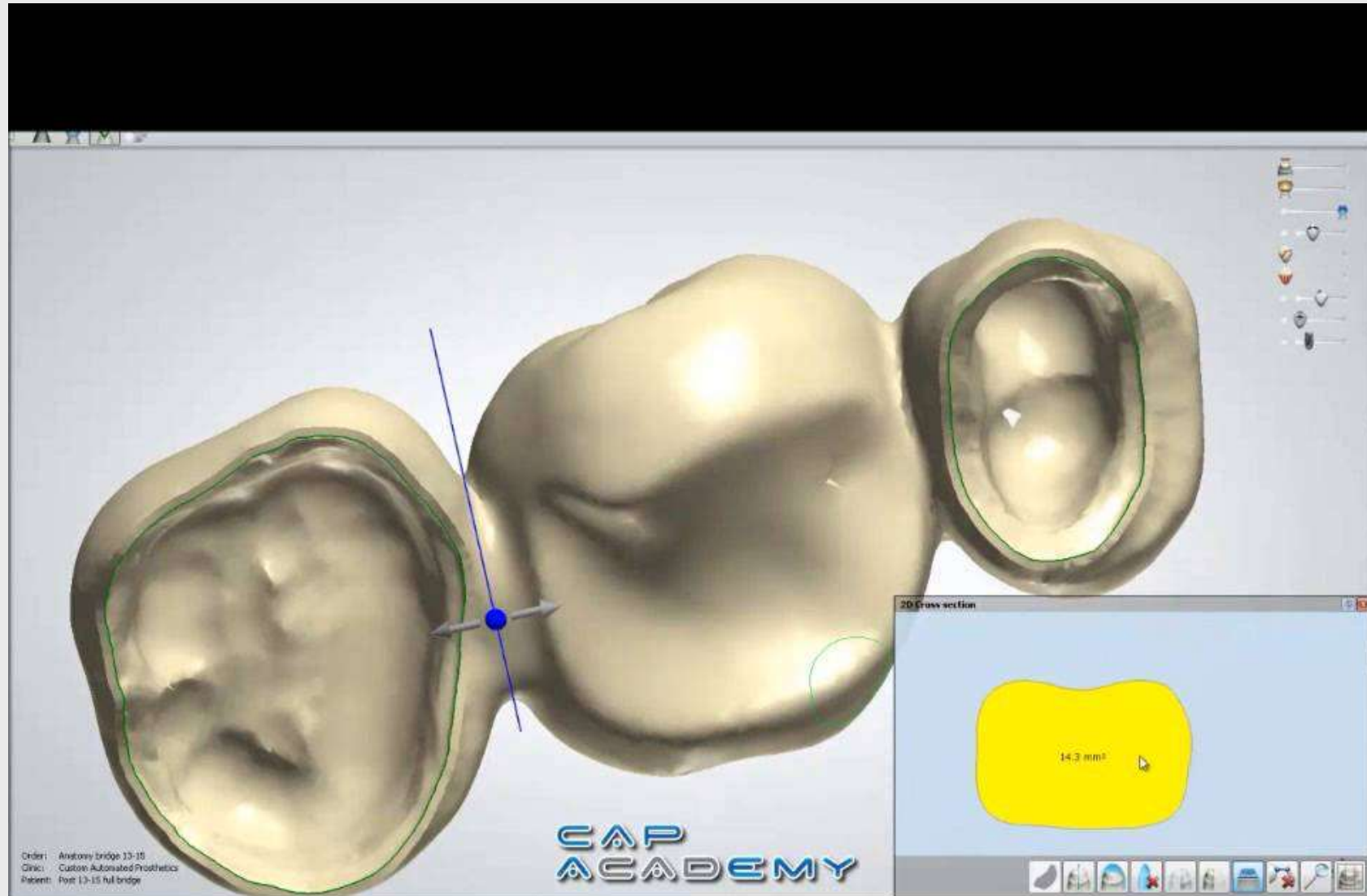
3- Connector:

It is that part of fixed partial denture that joins the individual components of the bridge together (the retainer and the pontic).

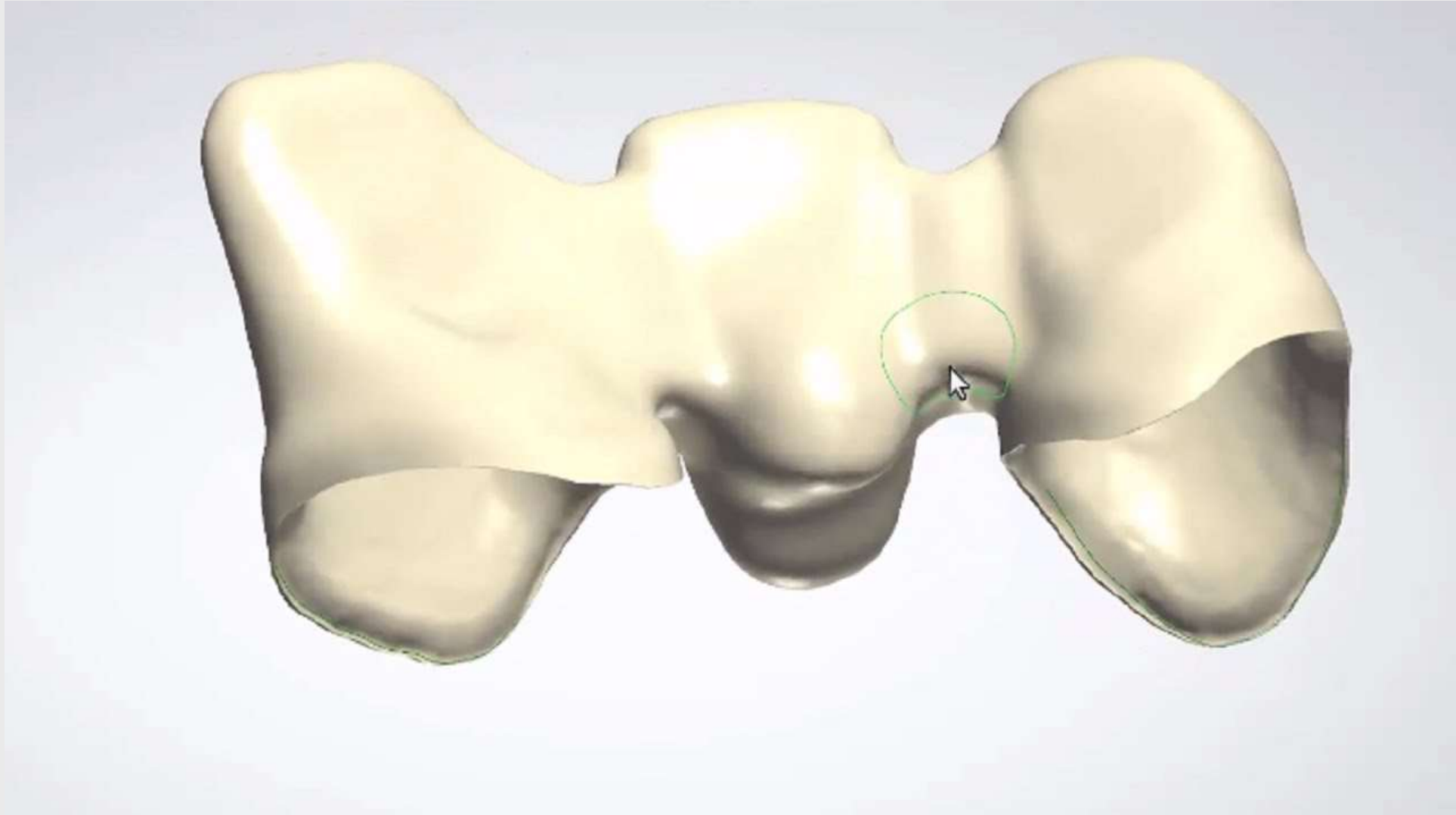
It could be fixed (rigid) or movable (flexible) connector.

When the retainer is attached to a fixed connector, it is called "major retainer", but when it is attached to a flexible (movable) connector it is called "minor retainer".

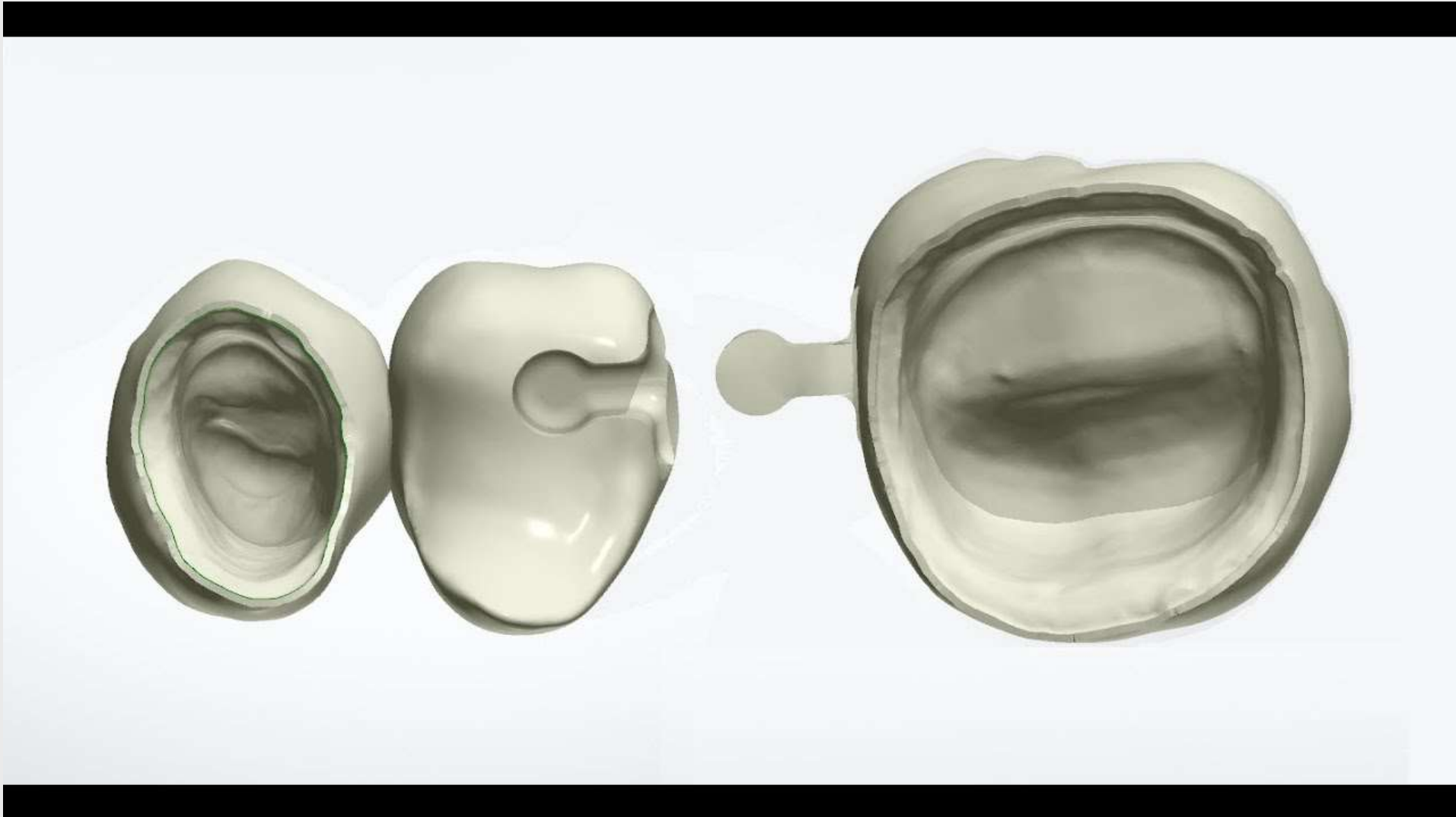
Rigid connectors



Rigid connectors



Movable connectors



Movable connectors For fixed-fixed bridge





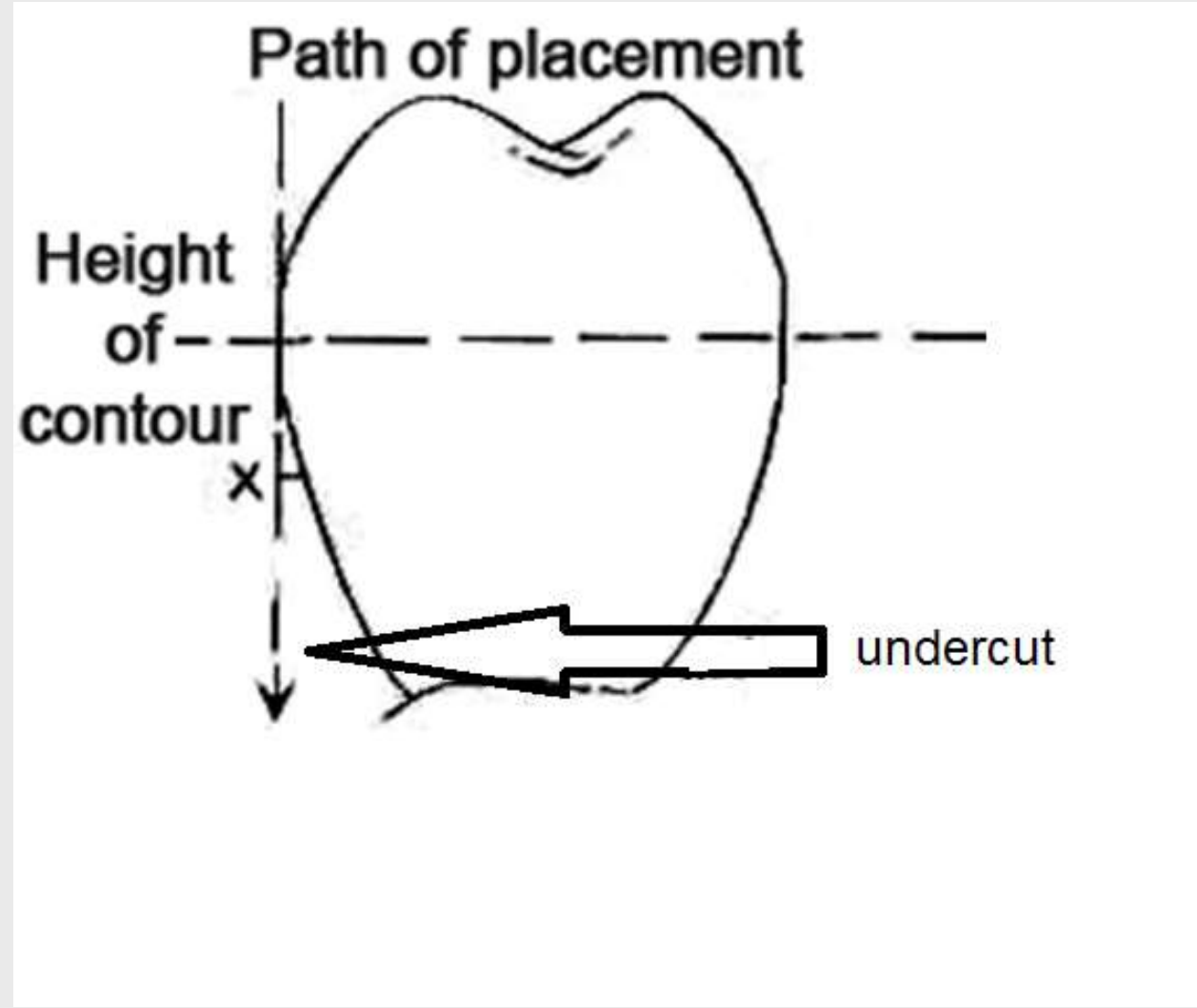
Movable connector for fixed removable bridge



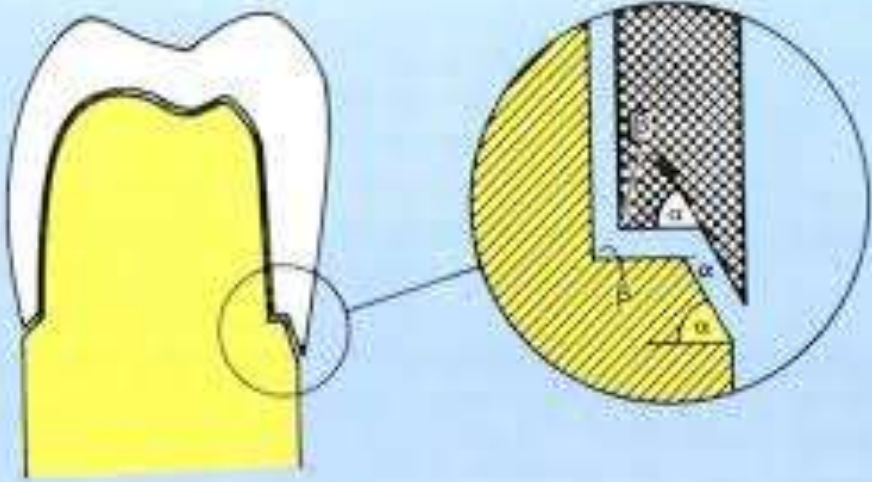
Purposes of crown construction:

- 1.To restore the grossly damaged tooth, fractured tooth or a tooth with a heavy filling [amalgam or composite].
- 2.To restore the masticatory function and speech.
- 3.To restore the esthetic [Improve the patients appearance].
- 4.To maintain the periodontal health by recontouring the occlusion and prevents food impaction.
- 5.As a retainer for the bridge.

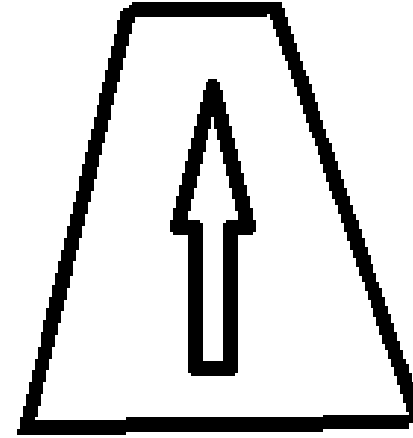
Under cut: it is the area below the height of contour on a surface. It may be desirable as in removable prosthesis or undesirable as in fixed prosthesis.



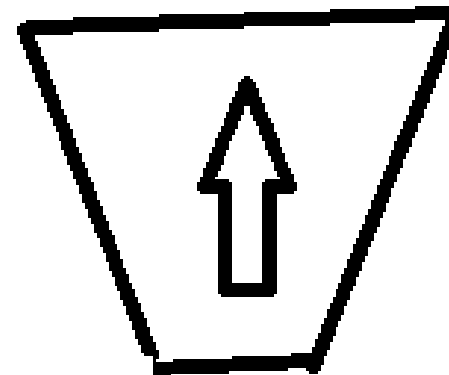
Finishing line: it is a junction which represent the end of tooth preparation or a junction between sound and prepared tooth structure.



Taper or slope: it is the relation between opposite walls of prepared tooth (converge or diverge).



converged



diverged

Any question

